



Theoretical proposal for a unified Bayesian model of adaptation in non-interactive and interactive speech production

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Abstract

In the scientific study of speech production, as well as in speech and language therapy, a variety of experimental paradigms and tasks resulting in plasticity and adaptation of the system are used. However, existing computational models accounting for adaptation appear segregated: feedback effects during non-interactive, altered speech are mostly accounted for by speech production models, whereas accommodation effects during interactive speech are mostly accounted for by speech perception models. In this paper, we consider COSMO, a previous unified Bayesian framework of speech perception and speech production, as the basis for a theoretical proposal and Bayesian model of adaptation effects. We show how these effects in both non-interactive and interactive speech production experiments could be described in this framework, thanks for its ability to separate knowledge and representations from their involvement during speech tasks.

Index Terms: speech production, modeling, perception-production

1. Introduction

The link between speech perception and production has been extensively studied, using a variety of experimental paradigms, including non-interactive speech (i.e., speech produced by a participant without any interlocutor) [1, 2, 3], or interactive speech such as dialogue [4, 5]. For a review, see [6].

In non-interactive speech production tasks, a common experimental manipulation consists in altering speech production, either through constraining the vocal tract itself (e.g., with a bite-block or lip-tube, or by stretching the skin of the cheeks), by altering somatosensory perception, or by manipulating the acoustic signal resulting from production (e.g., altered auditory feedback experiments). Such manipulations typically induce plasticity in the system, resulting in adaptation. In interactive speech production tasks, it is also common to observe plasticity and adaptation of the system: communication with an interlocutor, in and of itself, induces plasticity (e.g., phonetic convergence).

It is striking to note that, despite the apparent similarity of experimental situations and resulting plasticity, the theoretical and computational accounts of these phenomena appear mostly disjoint. On the one hand, non-interactive speech effects are typically explored by computational models of speech motor control [7, 8, 9, 10]. On the other hand, interactive speech effects are typically explored by computational models of speech perception [11, 12].

Of course, conceptual theories of the interaction between the speech perception and production systems have already been proposed, to integrate perception and comprehension in

a production conceptual model (e.g., Interactive Alignment theory [13]), or to consider the role of motor knowledge during speech perception (e.g., PACT [14]). Computational models of such theories would be useful, but are currently lacking.

Understanding the interplay between speech perception and production in non-interactive or interactive speech production contexts is also relevant from the point of view of speech therapy. Indeed, speech and language therapy often employs methods involving speaker model imitation and repetition, which engage both perception and production. Interestingly, some interventions for speech production disorders target both aspects simultaneously. For instance, the LSVT-LOUD method, widely used for speech production impairments in Parkinson's disease, exemplifies this approach by "retraining sensory perception of normal loudness (self-monitoring) and internal cueing (self-generating normal loudness)" [15, p. 1778]. In stuttering treatment, an effective fluency-enhancing technique involves listening and synchronizing with a metronome (e.g., [16]). This method requires producing each syllable or word in time with a regular auditory beat. Moreover, the interplay between perception and production is crucial when characterizing speech impairments. For instance, [17] suggested that a deficit in external auditory feedback monitoring in Parkinson's disease could lead to alterations in motor representations, ultimately resulting in production disorders.

Overall, developing an integrated computational model of speech perception and production is essential, not only for theoretical interest but also for better understanding and managing speech and language impairments. As of now, and to the best of our knowledge, no single integrative computational framework would be able to account for plasticity effects observed in the variety of experimental paradigms studying speech production.

In this paper, we propose to outline a theoretical proposal using Bayesian modeling to address this issue, paving the way towards a computational implementation. We consider the COSMO framework (Communicating Objects by Sensory-Motor Operations), a previous Bayesian model of speech perception and speech production [18, 19]. In the COSMO framework, each communicating agent is associated with a set of representations and knowledge, to relate their linguistic goals with articulatory and sensory spaces. The main assumption of the COSMO framework is that this limited set of representations and knowledge would be involved whatever the task at hand, that is, both during perception and production.

Therefore, our theoretical objective is to show how the COSMO framework could be applied to the variety of experimental situations relevant to the study of speech production: to account for adaptation effects during altered non-interactive speech, and to account for accommodation effects during interactive speech. In the rest of this paper, we first outline the main

effects of adaptation in non-interactive and interactive speech, along with the particular case of synchronous speech. We then recall the COSMO framework, and show how it would integrate these various observed phenomena in a unified conceptual framework.

2. Speech production experimental paradigms

2.1. Adaptation in non-interactive speech

One of the most common experimental contexts to study the interplay between speech perception and speech production involves a production task without interaction, and an alteration of the produced speech produced, for instance by modifying characteristics of the acoustic signal such as its pitch [10, 20], formant frequencies [2], or by adding delay [21]. The results from those experiments highlight that participants tend to compensate fully or partially for the perturbation they are exposed to. In theories of speech motor control, this can be explained by the effect of feedback on speech production [8]. The most common explanation of this phenomenon considers an auditory feedback mechanism, resulting from the difference between the expected and observed acoustic signals [7, 8, 22]. When such a difference occurs, it would be propagated back along the speech production system, so that the system would adapt, to correct future productions of the same utterance. In other words, this feedback-based adaptation system would be “internal” to the speech production system: the acoustic signal is only considered in its comparison with the expected acoustic consequence of production; no mapping of this acoustic signal into categories is involved.

Many computational models of speech motor control have already been proposed, including DIVA [8], TD [7], SFC [10], FACTS [9], ACT [23] and GEPPETO [24]. For a review, see [25]. Among these, some consider auditory-based feedback mechanisms, and can account for the adaptation responses in altered speech paradigms, at least qualitatively. These include DIVA [26, 8], FACTS [9], and SFC [10].

Importantly, an assumption generally made in this family of models is to consider that speech targets, once established through development by learning, would be invariant. Consequently, speech production adaptation would only help reach speech targets in adverse conditions [8, 22]. However, this strong delineation between learning and adaptation, while a viable first simplifying assumption, certainly is not straightforward. For instance, it is known that older community members unintentionally adjust their accents to reflect changes in speech patterns introduced by younger members of the community [27, 28].

To address that issue, the ACT model considers that feedback would stem from a high-level comparisons between abstract phoneme representations rather than actual auditory characteristics of speech sound ($F1$, $F2$, $F3$), inducing changes in auditory targets. In line with this hypothesis, a Bayesian model created to evaluate motor adaptation in speech production and perception [29] also suggested that simultaneous adaptation of auditory targets and of the sensorimotor component of the motor system would account for a variety of adaptation effects in altered speech experiments.

2.2. Adaptation in interactive speech

A more ecological situation than non-interactive speech is asynchronous interactive speech where participants communicate with an interlocutor, and alternate between speech comprehension and speech production (e.g., dialogue). Studies have been conducted where participants were interacting with a virtual agent [30] or with another human agent [11, 4]. In both situations, participants have been shown to accommodate their speech production features to characteristics of their interlocutor.

This phenomenon can be labeled as “speech accommodation” between speakers and is suggested to occur at different linguistic levels [4] and happen for phonetic details [30] as well as for speech rate and rhythm [31]. According to the Interactive Alignment model [13], covert imitation of speech is generated by both interlocutors, and speech is perceived through an inverse forward model of speech production. With the further assumption that the systems in speech production and perception are common, this leads to mutual accommodation between the interlocutors. However, in the Interactive Alignment model, accommodation for phonetic detail has not been explicitly introduced. In line with [6], we consider that it would still predict and account for phonetic accommodation.

Another theory on accommodation, the Communication Accommodation Theory [32] offers a complementary perspective by focusing on the social motivations behind speech adaptation. It suggests that speakers may converge or diverge in their speech patterns based on their desire to affiliate with or distance themselves from their interlocutors.

To our knowledge, there is no computational model of speech production that would be able to account for accommodation effects in interactive speech production. However, computational models of speech perception have been applied to account for accommodation effects. For instance, a Bayesian framework of speech perception [12] proposed that listeners adapt how they map auditory signals to phonetic units to their interlocutor’s mapping. A more recent Bayesian framework [11] also proposed different methods of adaptation for listeners to accommodate their phonetic mapping in a more constrained environment (listeners were asked to explicitly adapt to a speaker).

2.3. Adaptation in synchronous speech

Previously, we considered two situations of speech production that can induce adaptation: non-interactive speech, where adaptation is mostly interpreted as feedback-based, and interactive speech, where adaptation is mostly interpreted as accommodation-based. These situations also differ on the source of acoustic stimuli (endogenous in non-interactive speech; exogenous in interactive speech) and their temporal dimension (synchronous or almost synchronous with production in non-interactive speech; mostly asynchronous with production in interactive speech).

One can thus wonder what would happen in a situation where interlocutors would speak synchronously with one another, and whether both feedback-based and accommodation-based adaptation would occur. In an experiment involving both effects [5], participants were asked to speak synchronously with a recorded speaker. In this experiment, participants were shown to converge towards the $F0$ of the recorded speaker, but at least one fifth of the participants were shown to diverge from the $F0$ of the recorded speaker. In another synchronous speech experiment [33], all participants synchronized their speech to one an-

other, but only half of them had their first two formant frequencies shifted in response to altered speech perceived during the task. Results showed that participants who had their speech modified had similar response on average, but with increased variability.

These results suggest that feedback and accommodation could influence speech production adaptation simultaneously. The increased variability would originate from individual differences in the balance between both adaptation mechanisms. As we have argued above, computational models of speech production so far do not appear able to consider such combinations of adaptation mechanisms, since they rather focus specifically on one of them. To fully comprehend the interplay between various adaptation mechanisms, a comprehensive computational model is essential.

Outside of speech modeling, a computational model of rhythm production and adaptation in music was developed, that describes an analogous situation to synchronous speech, with concurrent rhythm production and listening. The Phase and Tempo Inference from Point Process Event Timing model [34] is a Bayesian model of rhythmic tracking in which tempo is continuously inferred based on prior expectations and sensory events. Rhythm estimates are dynamically adjusted by integrating event timing, uncertainty, and expectation-based distortions. This approach effectively captures key aspects of human rhythm perception and adaptation, particularly in the context of tapping with rhythmic cueing.

Synchronous speech suggests a final argument concerning the distinction, if it exists, between feedback-based adaptation and accommodation. Indeed, consider extreme cases of synchronous speech, characterized by dimensions including the temporal delay between speakers, the difference between vocal characteristics and between phonetic contents. First, imagine a situation without any temporal delay between speakers, without any timbre difference, and where speakers would pronounce exactly the same thing. This situation of “synchronous speech” would be perfectly indistinguishable from non-interactive speech, that is, there is a single speaker hearing the auditory consequence of his or her own production. Second, imagine the other extreme situation, with a large temporal mismatch, an obvious timbre difference between the two speakers, and difference in speech contents: this corresponds to “synchronous speech” with overlapping speech segments, as is commonly observed during dialogue.

If one extreme situation clearly elicits feedback-based adaptation, and the other accommodation, and if there exists a continuum between the situations, this strongly suggests that there also would be a continuum between adaptation mechanisms. Instead of hypothesizing a general mechanism that would “look like” feedback-based adaptation or accommodation in extreme cases, in the context of the current paper, our position is to, at least, consider that various combinations of feedback-based adaptation and accommodation should be integrated into a single computational model.

3. Theoretical proposition of a Bayesian framework

While traditional computational models of speech production excel at handling feedback-based adaptation in non-interactive speech, they struggle to account for the flexible nature of speech targets and accommodation phenomena observed in interactive contexts, which are mostly considered by computational models

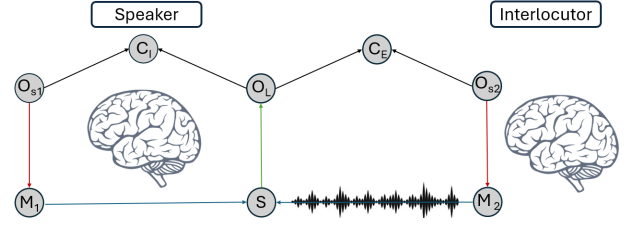


Figure 1: *Theoretical Bayesian model of speech production and perception. Adapted from [18]*

of speech perception. Moreover, we have argued that a continuum with varying degrees of combination between feedback-based adaptation and accommodation should be integrated in a computational model. Therefore, our theoretical objective is to propose a unified conceptual model, that would describe how, in speech motor control, the two seemingly distinct mechanisms of internal feedback-based adaptation and external accommodation could be reconciled. To do so, we first briefly review the Bayesian COSMO framework. Then, we will show how various adaptation effects could be accounted for, providing the first computational model taking into account adaptation in a unified perception-production framework.

3.1. The COSMO framework

The COSMO framework is a family of models that were developed as a unified computational model encompassing both production and perception tasks, and involving both a perceptual and a motor system [18]. In its initial and simpler form, COSMO is a Bayesian model relating linguistic objects O , which we assume to be phonemes in the context of this paper and for simplicity, articulatory gestures M and acoustic signals S . A probabilistic model involving these three representational spaces is defined by the joint probability distribution $P(C O_L S M O_S)$, decomposed as:

$$P(C O_L S M O_S) = P(O_S)P(M | O_S)P(S | M) P(O_L | S)P(C | O_L O_S). \quad (1)$$

This dependency architecture is represented Figure 1.

A central assumption of the COSMO framework is that representation spaces and the knowledge components that relate them are the same, whatever the considered task. This poses a slight terminological challenge to describe the terms appearing in Eq. (1), as the interpretation of model components are usually task or context dependent. To take a simple example relevant to the current paper: in speech production, S would typically refer to the acoustic consequence of a motor gesture, whereas in speech perception, S would refer to the acoustic stimulus. Nevertheless: $P(O_L)$ would be interpreted as a prior probability distribution over phonemes “from the point of view of the speaker”; $P(M | O_S)$ would be the phonemic motor repertoire associating motor gestures to phonemes; $P(S | M)$ would be an internal representation of the articulatory-to-acoustic transformation; $P(O_L | S)$ would be a phonemic sensory classifier; and finally, $P(C | O_L O_S)$, with C a Boolean variable, would assess or enforce an equality constraint on the two phonemic representations O_L and O_S .

In interactive situations, the communicating agent described by the $P(C O_L S M O_S)$ probability distribution, illustrated in Figure 1, interacts with the environment and an interlocutor. Therefore, we also consider the interlocutor’s phone-

mic objects (differentiating O_{S1} from O_{S2}), their motor gestures (differentiating M_1 and M_2), their acoustic consequences through the physical medium (M_2 to S), and an external constraint verification mechanism on the phonemic agreement between interlocutors (Boolean variable C_E , differentiated from the internal constraint C_I). In what follows, variables are indexed to differentiate between the speaker and the interlocutor when it is useful.

3.2. Bayesian accounts of the three speech production situations

In the COSMO framework, speech production corresponds to computing, from the joint probability distribution $P(C \ O_L \ S \ M \ O_S)$, a term of the form $P(M \mid O)$, that is, the probability distribution over motor gestures M given some intended linguistic object O [18]. The COSMO framework subsumes existing theories of speech production: if the considered linguistic target O is O_S , computing $P(M \mid O_S)$ amounts to a purely motor theory of speech production; if the target O is O_L , computing $P(M \mid O_L)$ amounts to a sensory theory of speech production (e.g., motor production to reach auditory targets if S is acoustic in nature); finally, if both linguistic object spaces are considered (through the constraint $C = \text{True}$), then computing $P(M \mid O_S \ [C = \text{True}])$ amounts to a perceptuo-motor theory of speech production.

In non-interactive speech, the first step of the process is endogenous production of a speech segment. For simplicity, we consider here a purely motor theory of speech production. Therefore, the first step is selecting a motor gesture m , for instance by drawing it at random according to $P(M_1 \mid O_{S1})$. Producing m has two consequences: first, acoustic consequences s through the physical environment; second, prediction of expected consequences s' through the internal model of the physical transformation.

In non-interactive speech, observing a discrepancy between s and s' suggests that some or all components of the system are not perfect, and could be improved. This is the starting point of any feedback-based adaptation mechanism. At this stage, most computational models assume that this would yield adaptation of either the internal model linking motor gestures to sensory consequences [22, 9] or in the classifiers associating acoustic signals with objects and motor gestures with objects [23]; while others argue that adaptation would affect both the auditory classifier and internal model [29].

In interactive speech, the first step of the process is receiving an acoustic signal s produced by an interlocutor. The main task in this context is speech comprehension. In the COSMO framework and assuming a perceptuo-motor theory of speech perception, this is performed by computing $P(O_{S1} \mid [S = s] \ [C_I = \text{True}])$. Technically, thanks to the constraint $[C_I = \text{True}]$, this amounts to inferring both O_{S1} and O_L from stimulus s . Moreover, computing this term also involves inference over the motor space M_1 , which can be interpreted as analysis-by-synthesis.

In interactive speech, accommodation effects suggest that this perception process, involving the stimulus s , inferred o , and implicitly inferred m , would yield adaptation of the components of the model. Existing models of accommodation would consider that adaptation occurs in the internal model linking motor gestures to sensory consequences [13], or an adaptation of the classifier linking acoustic signal and objects [12].

In synchronous speech, the process starts with the production of an acoustic signal s , a prediction of this signal s' , and the

simultaneous perception of a signal produced by an interlocutor s'' . The task in this context would be to both decode s'' , as in interactive situations, and monitor production by examining any difference between s and s' , as in non-interactive situations.

We postulate that the simultaneous performance of both situations would generate dependence, and possibly, interference. At the very least, they raise a “stimulus attribution” issue: if the voice of the interlocutor is not detected as such and is considered as coming from the speaker, it could trigger the feedback pathway and yield feedback-based adaptation; if the voice of the speaker is treated as external input, it could trigger the external pathway and yield accommodation. We hypothesize that the observed results in synchronous speech experiments, namely inter-individual differences [5, 33, 35], could be accounted for by variability in stimulus attribution. In the speech-to-speech synchronization experiment [35], participants could form expectations s' , but, because they did not hear their voice, they could not perceive s , and could use s'' instead. Our framework could explain the variability observed between participants, assuming that some would rely on s'' , more so than others. Manipulating the recorded speaker’s acoustic signal, by gradually making it closer or further away from the acoustic signal generated by participants, could be used to test this hypothesis.

4. Discussion

This paper examined whether a unified framework could account for both feedback-based and accommodation-based adaptation. To do so, we considered three experimental situations and the Bayesian COSMO model. Thanks to its close link between perception and production, we have shown that the COSMO model can effectively be a relevant framework. Importantly, we proposed a new hypothesis to explain individual differences in adaptation reported in the literature. Such a unified framework focusing on the the perception-production link and explaining inter-individual differences is also relevant for better understanding speech production impairments as well as developing intervention methods.

Our proposition has limitations. First, the motor variable M , at the level of detail of the COSMO framework, does not distinguish between motor planning and motor programming, two core concepts in motor control. For instance, in DIVA, adaptation to altered feedback does not change motor plans but motor programs. Furthermore, temporal aspects should be developed to tease apart the temporal dynamics of adaptation at differing linguistic time-scales, and to take into account the delay between prediction and observation that would elicit more feedback than accommodation. In addition, the sensory variable S should also be teased apart, to differentiate between acoustic and somatosensory dimensions first, as usually done in theories of motor control; and further, somatosensory information could be split into proprioceptive and tactile dimensions.

Our proposal should be implemented, and simulated as a computational model; its ability to account for non-interactive speech and interactive speech needs to be assessed. Our new hypothesis regarding the continuum between the two extreme situations provides testable predictions which should be evaluated in simulations. Manipulating the acoustic signal in a speech-to-speech synchronization task to make it more or less different from the speaker’s acoustic signal, be it in pitch, spectral content or rhythm, could be a first informative experiment to assess the existence of this continuum and predict the magnitude of adaptations.

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6. References

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